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Arthur Cayley

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[654] On Certain Octic Surfaces (concluded).

(Page 81.) In fact, the equation of the surface may be written in the form

$$w^8 + \left\{f^2x^4 + g^2y^4 + h^2z^4 - 2ghy^2z^2 - 2hfz^2x^2 - 2fgx^2y^2\right\} w^6$$
$$+2w^4 \left\{\begin{array}{l} - cfx^2y^2 - agy^2z^2 - bhz^2x^2 + 2kx^2y^2z^2 \\ + bfx^2z^2 + cgy^2x^2 + ahz^2y^2 \end{array}\right\}$$
$$+ \left(ay^2z^2 + bz^2x^2 + cx^2y^2\right)^2 = 0,$$

which puts in evidence the nodal curve

$$w = 0, \qquad -ay^2z^2 - bz^2x^2 - cx^2y^2 = 0 :$$

there are three similar forms which put in evidence the other three nodal curves.

The four curves are so related to each other that every line which meets three of them meets also the fourth curve; there is consequently a singly infinite series of lines meeting each of the four curves; these break up into four series of lines each forming an octic scroll, and each scroll has the four curves for nodal curves respectively; that is, each scroll is a surface included under the foregoing general equation, and derived from it by assigning a proper value to the constant k. To determine these values, write

$$\lambda + \mu + \nu = 0,$$
$$\frac{\lambda}{af} + \frac{\mu}{bg} + \frac{\nu}{ch} = 0,$$
$$\frac{k-\mu}{af} + \frac{\nu-\lambda}{bg} + \frac{\mu-\nu}{ch} = 0,$$

equations which give four systems of values for the ratios $(\lambda : \mu : \nu)$. We have then

$$k = af \cdot \frac{\nu-\mu}{\lambda} + bg \cdot \frac{\lambda-\nu}{\mu} + ch \cdot \frac{\mu-\lambda}{\nu},$$

viz. k has four values corresponding to the several values of $(\lambda : \mu : \nu)$.

The scroll in question is M. De La Gournerie's scroll Σ_1; the equation of the scroll Σ_1 is consequently obtained from the octic equation by writing therein the last-mentioned value of k.

It is to be noticed that k is, in effect, determined by a quartic equation; and, that, for a certain relation between the coefficients, this equation will have a twofold root. Assuming that this relation is satisfied, and assigning to k its twofold value, the resulting scroll becomes a torse; that is, two of the four scrolls coincide together and degenerate into a torse; corresponding to the remaining two values of k we have two scrolls, companions of the torse. In order to a twofold value of k, we must have

$$\frac{af}{\lambda^2} = \frac{bg}{\mu^2} = \frac{ch}{\nu^2},$$

and thence

$$(af)^{\frac{1}{3}} + (bg)^{\frac{1}{3}} + (ch)^{\frac{1}{3}} = 0;$$

or, what is the same thing,

$$(af + bg + ch)^3 - 27\,abcfgh = 0.$$

If for a moment we write $af = \alpha^3$, $bg = \beta^3$, $ch = \gamma^3$, and, therefore, $\alpha + \beta + \gamma = 0$; then for the twofold root, we have $\lambda : \mu : \nu = \alpha : \beta : \gamma$, and consequently

$$k = \alpha^2(\gamma - \beta) + \beta^2(\alpha - \gamma) + \gamma^2(\beta - \alpha) = (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha),$$

that is,

$$k = \left\{(af)^{\frac{1}{3}} - (bg)^{\frac{1}{3}}\right\}\left\{(bg)^{\frac{1}{3}} - (ch)^{\frac{1}{3}}\right\}\left\{(ch)^{\frac{1}{3}} - (af)^{\frac{1}{3}}\right\},$$

which agrees with the result in regard to the octic torse.

If in the octic equation we write (x, y, z, w) in place of (x^2, y^2, z^2, w^2), then we have the quartic equation

$$\begin{aligned}
&f^2x^2w^2 + g^2y^2w^2 + h^2z^2w^2\\
&+ 2hcx^2yz - 2cfx^2yw + 2bfx^2zw\\
&+ 2ahy^2zx - 2agy^2zw + 2cgy^2xw\\
&+ 2abz^2xy - 2bhz^2xw + 2ahz^2yw\\
&- 2ghw^2yz - 2hfw^2zx - 2fgw^2xy\\
&+ 2k\,xyzw = 0,
\end{aligned}$$

which is the equation of a quartic surface touched by the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$, in the four conics

$$\begin{aligned}
x &= 0, & hzw - gwy + ayz &= 0,\\
y &= 0, & -hzw + fwx + bzx &= 0,\\
z &= 0, & gyw - fwx + cxy &= 0,\\
w &= 0, & -ayz - bzx - cxy &= 0,
\end{aligned}$$

respectively.

II. The octic surface

$$\begin{aligned}
U = {} & b^2c^2f^2x^4 + c^2a^2g^2y^4 + a^2b^2h^2z^4 + f^2g^2h^2w^4\\
& - 2a^2cg(bg - ch)\,y^2z^2 - 2b^2ah(ch - af)\,z^2x^2 - 2c^2bf(af - bg)\,x^2y^2\\
& + 2a^2bh(\,\cdot\,)\,y^2z^2 + 2b^2cf(\,\cdot\,)\,z^2x^2 + 2c^2ag(\,\cdot\,)\,x^2y^2\\
& - 2f^2bc(\,\cdot\,)\,x^2w^2 - 2g^2ca(\,\cdot\,)\,y^2w^2 - 2h^2ab(\,\cdot\,)\,z^2w^2\\
& + 2f^2gh(\,\cdot\,)\,x^2w^2 + 2g^2hf(\,\cdot\,)\,y^2w^2 + 2h^2fg(\,\cdot\,)\,z^2w^2\\
& + f^2(b^2g^2 + c^2h^2 - 4bgch)\,w^2x^2 + g^2(c^2h^2 + a^2f^2 - 4chaf)\,w^2y^2 + h^2(a^2f^2 + b^2g^2 - 4afbg)\,w^2z^2\\
& + a^2(\,\cdot\,)\,y^2z^2 + b^2(\,\cdot\,)\,z^2x^2 + c^2(\,\cdot\,)\,x^2y^2\\
& - 2gh(bcgh - a^2f^2 - 2afbg - 2afch)\,w^2y^2z^2\\
& - 2bh(\,\cdot\,)\,z^2x^2\\
& + 2cg(\,\cdot\,)\,y^2x^2w^2\\
& + 2bc(\,\cdot\,)\,x^2y^2z^2
\end{aligned}$$

$$
\begin{aligned}
&- 2hf(cahf - b^2g^2 - 2bgaf - 2bgch)\, w^2z^2x^2 \\
&- 2cf(\,\cdot\,)\, x^2y^2 \\
&+ 2ah(\,\cdot\,)\, z^2y^2w^2 \\
&+ 2ca(\,\cdot\,)\, y^2x^2z^2 \\
&- 2fg(abfg - c^2h^2 - 2chaf - 2chbg)\, w^2x^2y^2 \\
&+ 2ag(\,\cdot\,)\, y^2z^2w^2 \\
&+ 2bf(\,\cdot\,)\, x^2z^2w^2 \\
&+ 2ab(\,\cdot\,)\, z^2x^2y^2 \\
&+ 2\Omega\, x^2y^2z^2w^2 = 0,
\end{aligned}
$$

where the values of the coefficients indicated by $(\,\cdot\,)$ are at once obtained by the proper interchanges of the letters, and where Ω is an arbitrary coefficient, is a surface having the four nodal conics

$$
\begin{aligned}
x &= 0, & cy^2 - bz^2 + fw^2 &= 0, \\
y &= 0, & -cx^2 + az^2 + gw^2 &= 0, \\
z &= 0, & bx^2 - ay^2 + hw^2 &= 0, \\
w &= 0, & -fx^2 - gy^2 - hz^2 &= 0.
\end{aligned}
$$

In fact, writing the equation under the form

$$w^4\Theta + (fx^2+gy^2+hz^2)^2 \times (b^2c^2x^4+c^2a^2y^4+a^2b^2z^4-2a^2bcy^2z^2-2b^2caz^2x^2-2c^2abx^2y^2) = 0,$$

we put in evidence the nodal conic $w = 0$, $fx^2+gy^2+hz^2 = 0$: and similarly for the other nodal conics.

It is to be observed, that the complete section by the plane $w = 0$ is the conic $fx^2+gy^2+hz^2 = 0$, twice repeated, and the quartic

$$b^2c^2x^4 + c^2a^2y^4 + a^2b^2z^4 - 2a^2bcy^2z^2 - 2ab^2cz^2x^2 - 2abc^2x^2y^2 = 0:$$

the latter being the system of four lines

$$\frac{x}{\sqrt{(a)}} \pm \frac{y}{\sqrt{(b)}} \pm \frac{z}{\sqrt{(c)}} = 0.$$

The plane in question, $w = 0$, meets the other nodal conics in the six points

$$(x = 0,\ bz^2 - cy^2 = 0), \quad (y = 0,\ cx^2 - az^2 = 0), \quad (z = 0,\ ay^2 - bx^2 = 0),$$

which six points are the angles of the quadrilateral formed by the above-mentioned four lines.

The four conics are such, that every line meeting three of these conics meets also the fourth conic. The lines in question form a double system: each of these systems has, in reference to any pair of nodal conics, a homographic property as follows: viz. considering for example the two conics in the planes $z = 0$ and $w = 0$ respectively, if a line meets these conics in the points P and Q respectively, and through these points respectively and the line $x = 0$, $y = 0$ we draw planes, then the system of the P planes and the system of the Q planes correspond homographically to each other, the coincident planes of the two systems being the planes $x = 0$ and $y = 0$ respectively.

Conversely, if through the line $(x = 0, y = 0)$ we draw the two homographically related planes meeting the two conics in the points P and Q respectively, then, for a proper value (determined by a quadratic equation) of the constant $k\,(= \theta' \div \theta)$ which determines the homographic relation, the line PQ will be a line meeting each of the four conics, and will belong to one or other of the above-mentioned two systems, as k is equal to one or the other of the two roots of the quadratic equation. The scroll generated by the lines meeting each of the four conics, or what is the same thing, any three of these conics, is *primâ facie* a scroll of the order 16; but by what precedes, it appears that this scroll breaks up into two scrolls, which will be each of the order 8. Moreover, each scroll has the four conics for nodal curves; and since the equation $U = 0$ is the general equation of an octic surface having the four conics for nodal curves, it follows, that the equation of the scroll is derived from that of the octic surface $U = 0$, by assigning a proper value to the indeterminate coefficient Ω; so that there are in fact two values of Ω, for each of which the surface $U = 0$ becomes a scroll.

To sustain the foregoing conclusions, take $x = \theta y$, $x = \theta' y$ for the equations of the two planes through the line $(x = 0, y = 0)$, which meet the z-conic and w-conic in the points P and Q respectively; then the equations of the line PQ are

$$\sqrt{(f\theta^2 + g)}\,(x - \theta' y) + \sqrt{(-h)}\,(\theta' - \theta)\,z = 0,$$

$$-\sqrt{(b\theta'^2 - a)}\,(x - \theta y) + \sqrt{(-h)}\,(\theta' - \theta)\,w = 0,$$

or, writing therein $\theta' = k\theta$, the equations are

$$\sqrt{(f\theta^2 + g)}\,(x - k\theta y) + \sqrt{(-h)}\,(k - 1)\,\theta z = 0,$$

$$-\sqrt{(bk^2\theta^2 - a)}\,(x - \theta y) + \sqrt{(-h)}\,(k - 1)\,\theta w = 0.$$

To find where the line in question meets the plane $y = 0$, we have

$$\sqrt{(f\theta^2 + g)}\,x + \sqrt{(-h)}\,(k - 1)\,\theta z = 0,$$

$$-\sqrt{(bk^2\theta^2 - a)}\,x + \sqrt{(-h)}\,(k - 1)\,\theta w = 0,$$

and thence

$$(f\theta^2 + g)\,x^2 + h(k - 1)^2\theta^2 z^2 = 0,$$
$$(bk^2\theta^2 - a)\,x^2 + h(k - 1)^2\theta^2 w^2 = 0,$$

or multiplying by a, g and adding

$$(af + bgk^2)\,x^2 + h(k - 1)^2(az^2 + gw^2) = 0,$$

or assuming

$$af + bgk^2 + ch(k - 1)^2 = 0,$$

the equation is

$$-cx^2 + az^2 + gw^2 = 0.$$

That is, k being determined by the quadric equation $af + bgk^2 + ch(k-1)^2 = 0$, the line PQ meets the y-conic $y = 0$, $-cx^2 + az^2 + gw^2 = 0$; and, in a similar manner, it appears that the line PQ also meets the x-conic $x = 0$, $cy^2 - bz^2 + fw^2 = 0$.

Writing for greater symmetry $1 : -k : k-1 = \lambda : \mu : \nu$, we have

$$\lambda + \mu + \nu = 0,$$

$$af\lambda^2 + bg\mu^2 + ch\nu^2 = 0,$$

so that there are two systems of values of (λ, μ, ν) corresponding to, and which may be used in place of, the two values of k respectively.

Starting now from the equations

$$(f\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2\theta^2 z^2 = 0,$$

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2 w^2 = 0,$$

the elimination of θ from these equations leads to an equation $U = 0$, of the above-mentioned form but with a determinate value of the coefficient.

The process, although a long one, is interesting and I give it in some detail.

Elimination of θ from the foregoing equations.

We have

$$U = M\,\Pi\left[(f\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2\theta^2 z^2\right],$$

where Π denotes the product of the expressions corresponding to the four roots of the equation

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2 w^2 = 0.$$

Observing that this equation does not contain z, and that the expression under the sign Π does not contain w, it is at once seen that the product Π is in regard to (z, w) a rational and integral function of the form $(z^2, w^2)^4$; and since, in regard to (z, w), U is also a rational and integral function of the same form $(z^2, w^2)^4$, it is clear that the factor M does not contain z or w, but is a function of only (x, y). To determine it we may write $z = 0$, $w = 0$: this gives

$$c^4(bk^2\theta^2 - ay^2)^4(fx^2 + gy^2)^4 = M\,h^4(f\theta^2 + g)^4(k\theta y - x)^8,$$

where

$$(bk^2\theta^2 - a)(\theta y - x)^2 = 0,$$

and the values of θ are therefore $+\dfrac{x}{y}$, $-\dfrac{x}{y}$, $+\dfrac{\sqrt{(a)}}{k\sqrt{(b)}}$, $-\dfrac{\sqrt{(a)}}{k\sqrt{(b)}}$. Hence substituting and observing that

$$ch^2(k-1)^2 = (af + bgk^2)^2,$$

it is easy to find

$$M = \frac{x^4}{y^4} \cdot \frac{h^4(k-1)^2}{b^4k^4},$$

that is, we have

$$U = \frac{x^4}{y^4} \cdot \frac{h^4(k-1)^2}{b^4k^4} \, \Pi\left[(f\theta^2+g)(k\theta y - x)^2 + h(k-1)^2\theta^2 z^2\right],$$

where

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2 w^2 = 0.$$

If for greater convenience we write $\theta = \dfrac{x}{y}\phi$, then this formula becomes

$$U = \frac{x^4}{y^4} \cdot \frac{h^4(k-1)^2}{b^4k^4} \, \Pi\left[(fx^2\phi^2+gy^2)(k\phi - 1)^2 + h(k-1)^2\phi^2 z^2\right],$$

where

$$(bk^2x^2\phi^2 - ay^2)(\phi - 1)^2 + h(k-1)^2\phi^2 w^2 = 0,$$

or, what is the same thing,

$$(bk^2x^2\phi^2 - ay^2)(\phi - 1)^2 + h(k-1)^2\phi^2 w^2 = 0.$$

Suppose that the terms in U which contain z^2 are $= \Theta z^2$; then we have

$$\Theta z^2 = \frac{x^4}{y^4} \cdot \frac{h^4(k-1)^2}{b^4k^4} \, \Sigma'\left[\Pi'\left\{(fx^2\phi^2+gy^2)(k\phi-1)^2\right\} \cdot h(k-1)^2\phi^2 z^2\right],$$

or, what is the same thing,

$$\Theta = \frac{x^4}{y^4} \cdot \frac{h^5(k-1)^4}{b^4k^4} \, \Sigma\left[\phi^2\,\Pi'\left\{(fx^2\phi^2+gy^2)(k\phi-1)^2\right\}\right],$$

where Π' refers to the remaining three roots ϕ_2, ϕ_3, ϕ_4; this may also be written

$$\Theta = \frac{x^4}{y^4} \cdot \frac{h^5(k-1)^4}{b^4k^4} \, \Sigma\left\{\frac{\phi^2}{(fx^2\phi^2+gy^2)(k\phi-1)^2}\right\} \cdot \Pi\left\{(fx^2\phi^2+gy^2)(k\phi-1)^2\right\}.$$

Hence, observing that we have identically

$$fx^2\phi^2 + gy^2 = \{\phi x\sqrt{(f)} + iy\sqrt{(g)}\}\{\phi x\sqrt{(f)} - iy\sqrt{(g)}\},$$

and writing $\phi = \pm\dfrac{iy\sqrt{(g)}}{x\sqrt{(f)}}$, $\{i = \sqrt{(-1)}$ as usual$\}$, we find

$$\Pi\{\phi x\sqrt{(f)} \pm iy\sqrt{(g)}\} = -\frac{1}{b^2k^4x^4}\left[c\{x\sqrt{(f)} \pm iy\sqrt{(g)}\}^2 - fgv^2\right]y^2,$$

$$\Pi(k\phi - 1) = \frac{1}{k^2b^2x^4}(bx^2 - ay^2 + hw^2);$$

whence, writing for shortness

$$\begin{aligned}\Delta = &\left[c\{x\sqrt{(f)} + iy\sqrt{(g)}\}^2 - fgw^2\right]\left[c\{x\sqrt{(f)} - iy\sqrt{(g)}\}^2 - fgw^2\right]\\ &= c^2f^2x^4 + c^2g^2y^4 + f^2g^2w^4 + 2cfg^2y^2w^2 - 2cfgx^2w^2 + 2c^2fgx^2y^2,\end{aligned}$$

we find

$$\Pi(fx^2\phi^2+gy^2)=\frac{h^2(k-1)^4}{b^4k^8}\cdot\frac{y^4}{x^8}\,\Delta,$$

$$\Pi(k\phi-1)^2=\frac{1}{b^4k^8x^8}(bx^2-ay^2+hw^2)^2,$$

and thence

$$\Pi(fx^2\phi^2+gy^2)(k\phi-1)^2=\frac{h^2(k-1)^4}{b^8k^{16}}\,\Delta\,(bx^2-ay^2+hw^2)^2\,\frac{y^4}{x^{16}},$$

and consequently

$$\Theta=\frac{h^7(k-1)^8}{b^{12}k^{20}}\cdot\frac{1}{x^{12}}\,\Delta\,(bx^2-ay^2+hw^2)^2\,\Sigma\left\{\frac{\phi^2}{(fx^2\phi^2+gy^2)(k\phi-1)^2}\right\}.$$

Hence, writing

$$\Sigma\left\{\frac{\phi^2}{(fx^2\phi^2+gy^2)(k\phi-1)^2}\right\}=\frac{A}{(k\phi-1)^2}+\frac{B}{k\phi-1}+\frac{C}{x\phi\sqrt{(f)}+iy\sqrt{(g)}}+\frac{D}{x\phi\sqrt{(f)}-iy\sqrt{(g)}},$$

we may calculate separately the terms

$$\Sigma\left\{\frac{A}{(k\phi-1)^2}+\frac{B}{k\phi-1}\right\}$$

and

$$\Sigma\left\{\frac{C}{x\phi\sqrt{(f)}+iy\sqrt{(g)}}+\frac{D}{x\phi\sqrt{(f)}-iy\sqrt{(g)}}\right\}.$$

The first of these is

$$=\frac{1}{h(k-1)^2(fx^2+k^2gy^2)^2}\,(bx^2-ay^2+hw^2)^2\cdot\{x,y,w\}^2,$$

if for shortness

$$\begin{aligned}\{x,y,w\}^2=(fx^2+k^2gy^2)\big[&4\{(2-k)bx^2-ay^2+h(1-k)w^2\}^2\\ &-2(bx^2-ay^2+hw^2)\{(6-6k+k^2)bx^2-ay^2+h(1-k)^2w^2\}\big]\\ &+4k^2(k-1)gy^2-(bx^2-ay^2+hw^2)\{(2-k)bx^2-ay^2+h(1-k)w^2\};\end{aligned}$$

the second is

$$=\frac{2}{(k-1)^2(fx^2+k^2gy^2)^2}\cdot(x,y,w)^2,$$

if for shortness

$$\begin{aligned}(x,y,w)^2=\;&\big[(fx^2-k^2gy^2)(cfx^2-cgy^2-fgw^2)-4ckfgx^2y^2\big]\\ &\times\big[fgbk^2x^2+\{2ch(k-1)^2-af\}gy^2+fgh(k-1)^2w^2\big]\\ &+2\big[k^2bg-ch(k-1)^2\big]\,fgx^2y^2\big[c(k+1)(fx^2-kgy^2)-kfgw^2\big]\,;\end{aligned}$$

and hence

$$\Theta=\frac{h^6(k-1)^6}{b^{12}k^{20}(fx^2+k^2gy^2)^2}\,\Delta\big[\{x,y,w\}^2+2(bx^2-ay^2+hw^2)^2\,(x,y,w)^2\big]\,,$$

which must be a rational and integral function of (x,y,w).

In partial verification of this, observe that, because U contains the terms

$$2h^2cf(ch-af)\,x^2z^2+2\Omega\,x^2y^2z^2w^2,$$

Θ should contain the terms

$$2h^2cf(ch-af)\,x^2 + 2\Omega\, x^2y^2w^2,$$

viz. in the term in x^4 should be $= 2h^2cf(ch-af)\,x^4$.

Now writing $y=0$, $w=0$, we have

$$\Delta = c^2f^2x^4,$$

$$\{x,y,w\}^2 = bx^2\{4(2-k)^2 - 2(6-6k+k^2)\}\,x^4 = bx^2(4-4k+2k^2)\,x^4,$$

$$(x,y,w)^2 = bcfgk^2x^4;$$

and hence the required term of Θ is x^4 multiplied by

$$b^3c^3fh(4-4k+2k^2) + 2hb^2cfgk^2,$$

viz. the coefficient is

$$= 2b^2cf\left[c^2h(2-2k+k^2) + bgk^2\right] = 2b^2cf\left[ch + ch(1-k)^2 + bgk^2\right],$$

which in virtue of the relation $af + bgk^2 + ch(1-k)^2 = 0$ becomes, as it should do,

$$= 2b^2cf(ch-af).$$

The actual division by $(fx^2+k^2gy^2)^2$ would, however, be a very tedious process, and it is to be observed, that we only require to know the term $2\Omega x^2y^2w^2$ of Θ. We may therefore adopt a more simple course as follows: the terms of Θ which contain w^2 are $= (Ax^2 + 2\Omega x^2y^2 + By^4)\,w^2$, hence writing for a moment

$$\{x,y,w\}^2 = P + Qw^2, \qquad (x,y,w)^2 = R + Sw^2,$$

and observing that we have

$$\Delta = c^2(fx^2+gy^2)^2 - 2cfg(fx^2-gy^2)\,w^2 + \text{\&c.},$$

$$(bx^2-ay^2+hw^2)^2 = (bx^2-ay^2)^2 + 2h(bx^2-ay^2)\,w^2 + \text{\&c.},$$

we have

$$(fx^2+k^2gy^2)^2(Ax^2+2\Omega x^2y^2+By^4) = c^2h(fx^2+gy^2)^2Q - 2c^2fgh(fx^2-gy^2)P + (bx^2-ay^2)^2S + 2h(bx^2-ay^2)R.$$

But in this identical equation we may write $x^2=a$, $y^2=b$, which gives

$$(af+k^2bg)^2(Aa^2+2\Omega ab+Bb^2) = c^2h(af+bg)^2Q - 2c^2fgh(af-bg)P;$$

and from the equation

$$\{x,y,w\}^2 = P + Qw^2,$$

we have

$$\begin{aligned}
P + Qw^2 = {} & (af+bgk^2)\big[4\{(2-k)ab - ab + (1-k)hw^2\}^2 \\
& -2(ab+hw^2)\{(6-6k+k^2)ab - ab + (1-k)^2hw^2\}\big] \\
& + 4k^2(k-1)bghw^2(1-k)ab \\
= {} & -ch(k-1)^2\big[4(k-1)^2(a^2b^2+2w^2abh) - 2(6-6k+k^2)hw^2\big] \\
& - 4k^2(k-1)^2ab^2ghw^2,
\end{aligned}$$

that is,

$$Q = (k-1)^2 abh\big[ch(-6k^2+4k+2) - 4k^2bg\big]\,,$$

whence

$$(k-1)^2(Aa^2+2\Omega ab+Bb^2) = ab(af+bg)^2\big[ch(-6k^2+4k+2) - 4k^2bg\big] + 8fgh(af-bg)(k-1)^2(ab)^2.$$

But we have

$$Aa^2 + Bb^2 = -2ab(af-bg)(-afbg + c^2h^2 + 2chaf + 2chbg),$$

and thence

$$2(k-1)^2\Omega = (af+bg)^2\big[ch(-6k^2+4k+2) - 4k^2bg\big] + (k-1)^2(af-bg)\big[-2afbg + 2c^2h^2 + 4chaf + 4chbg\big]\,,$$

or

$$(k-1)^2\Omega = (af+bg)^2\big[ch(-3k^2+2k+1) - 2k^2bg\big] + (k-1)^2(af-bg)\big[3afbg + c^2h^2 + 2chaf + 2chbg\big]\,.$$

Writing $-3k^2+2k+1 = -3(k-1)^2 - 4(k-1)$, this is

$$\Omega = (af+bg)^2\left[ch\left(-3 - \frac{4}{k-1}\right) - 2\frac{k^2}{(k-1)^2}\,bg\right] + (af-bg)\big[3afbg + c^2h^2 + 2chaf + 2chbg\big]\,,$$

or since $1 : -k : k-1 = \lambda : \mu : \nu$; and writing for shortness $(af, bg, ch) = (\alpha, \beta, \gamma)$, this is

$$\Omega = (\alpha+\beta)^2\left(-3\gamma - \frac{4\gamma\lambda}{\nu} - \frac{2\mu^2}{\nu^2}\beta\right) + (\alpha-\beta)\{3\alpha\beta + \gamma^2 + 2\gamma\alpha + 2\gamma\beta\},$$

which is the value of Ω: viz. the conclusion arrived at is that, eliminating θ from the equations

$$(f\theta^2+g)(k\theta y - x)^2 + h(k-1)^2\theta^2z^2 = 0,$$

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2w^2 = 0,$$

where k denotes a determinate function of af, bg, ch, viz. writing af, bg, ch $= \alpha, \beta, \gamma$ and $1 : -k : k-1 = \lambda : \mu : \nu$, we have

$$\lambda + \mu + \nu = 0,$$

$$\alpha\lambda^2 + \beta\mu^2 + \gamma\nu^2 = 0,$$

equations which serve to determine k: the result of the elimination is the octic equation

$$b^2c^2f^2x^4 + \cdots + 2\Omega\, xyz^2w^2 = 0,$$

where Ω has the last-mentioned value.

The value of Ω is unsymmetrical in its form, and there are apparently six values; viz. writing

$$A = (\beta+\gamma)^2\left\{\alpha\left(-3-\frac{4\mu}{\nu}\right)-\frac{2\mu^2}{\nu^2}\gamma\right\}+(\beta-\gamma)(S+\beta\gamma+\alpha^2),$$

$$B = (\gamma+\alpha)^2\left\{\beta\left(-3-\frac{4\nu}{\lambda}\right)-\frac{2\nu^2}{\lambda^2}\alpha\right\}+(\gamma-\alpha)(S+\gamma\alpha+\beta^2),$$

$$C = (\alpha+\beta)^2\left\{\gamma\left(-3-\frac{4\lambda}{\mu}\right)-\frac{2\lambda^2}{\mu^2}\beta\right\}+(\alpha-\beta)(S+\alpha\beta+\gamma^2),$$

$$A_1 = (\beta+\gamma)^2\left\{\alpha\left(-3-\frac{4\nu}{\mu}\right)-\frac{2\nu^2}{\mu^2}\beta\right\}-(\beta-\gamma)(S+\beta\gamma+\alpha^2),$$

$$B_1 = (\gamma+\alpha)^2\left\{\beta\left(-3-\frac{4\lambda}{\nu}\right)-\frac{2\lambda^2}{\nu^2}\gamma\right\}-(\gamma-\alpha)(S+\gamma\alpha+\beta^2),$$

$$C_1 = (\alpha+\beta)^2\left\{\gamma\left(-3-\frac{4\mu}{\lambda}\right)-\frac{2\mu^2}{\lambda^2}\alpha\right\}-(\alpha-\beta)(S+\alpha\beta+\gamma^2),$$

where for shortness $S = 2(\beta\gamma+\gamma\alpha+\alpha\beta)$, the six values would be A, B, C, A_1, B_1, C_1.

But we have really

$$A = B = C = -A_1 = -B_1 = -C_1,$$

so that Ω has really only two values, equal and of opposite signs, or, what is the same thing, Ω^2 has a unique value. In fact, writing for shortness

$$\lambda+\mu+\nu = P, \qquad \alpha\lambda^2+\beta\mu^2+\gamma\nu^2 = X,$$

we find at once the identity

$$\lambda^2\nu^2(A+A_1) = (\beta+\gamma)^2(-2X-4X\alpha P),$$

so that $A = -A_1$, in value of $P = 0$, $X = 0$. And similarly $B = -B_1$, $C = -C_1$.

But the demonstration of the equation $A = B$ is more complicated. We have

$$\begin{aligned} A - B = & -3\alpha(\beta+\gamma)^2 - 4\alpha(\beta+\gamma)^2\frac{\mu}{\nu} - 2\gamma(\beta+\gamma)^2\frac{\mu^2}{\nu^2} + (\beta-\gamma)(S+\beta\gamma+\alpha^2) \\ & + 3\beta(\gamma+\alpha)^2 + 4\beta(\gamma+\alpha)^2\frac{\nu}{\lambda} + 2\alpha(\gamma+\alpha)^2\frac{\nu^2}{\lambda^2} - (\gamma-\alpha)(S+\gamma\alpha+\beta^2), \end{aligned}$$

that is,

$$\begin{aligned} \lambda^2\nu^2(A-B) = & \left\{-3\alpha(\beta+\gamma)^2 + 3\beta(\gamma+\alpha)^2 + (\beta-\gamma)(S+\beta\gamma+\alpha^2) - (\gamma-\alpha)(S+\gamma\alpha+\beta^2)\right\}\lambda^2\nu^2 \\ & - 4\alpha(\beta+\gamma)^2\lambda^2\mu\nu \\ & - 2\gamma(\beta+\gamma)^2\lambda^2\mu^2 \\ & + 4\beta(\gamma+\alpha)^2\nu^3\lambda \\ & + 2\alpha(\gamma+\alpha)^2\nu^4, \end{aligned}$$

or, denoting for a moment the coefficient of $\lambda^2\nu^2$ by K, and writing also $\gamma\nu^2 = X - \alpha\lambda^2 - \beta\mu^2$, $\nu = P - \lambda - \mu$, this is

$$\begin{aligned} &= K\lambda^2\nu^2 \\ &\quad - 4\alpha(\beta+\gamma)^2\lambda^2\mu\nu \\ &\quad - 2(\beta+\gamma)^2\mu^2(X - \alpha\lambda^2 - \beta\mu^2) \\ &\quad + 4\beta(\gamma+\alpha)^2\lambda(P-\lambda-\mu)\nu^2 \\ &\quad + 2\alpha(\gamma+\alpha)^2\nu^4, \end{aligned}$$

$$\begin{aligned} = \; & - 2(\beta+\gamma)^2\mu^2X + 4\beta(\gamma+\alpha)^2\lambda\nu^2P \\ & + 2\alpha(\gamma+\alpha)^2\nu^4 \\ & - 4\beta(\gamma+\alpha)^2\lambda^2\nu^2 \\ & + \{-4\beta(\gamma+\alpha)^2 + K + 2\alpha(\beta+\gamma)^2\}\lambda^2\nu^2 \\ & - 4\alpha(\beta+\gamma)^2\lambda^2\mu\nu \\ & + 2\beta(\beta+\gamma)^2\mu^4, \end{aligned}$$

and here the coefficient of $\lambda^2\nu^2$ is found to be

$$= 2\{\alpha\beta(\alpha+\beta) + \gamma(\alpha-\beta)^2 - 3\gamma^2(\alpha+\beta)\}.$$

Hence, the terms without X or P are $= 2V$, where

$$\begin{aligned} V = \; & \alpha(\gamma+\alpha)^2\nu^4 \\ & - 2\beta(\gamma+\alpha)^2\lambda^2\nu^2 \\ & + \{\alpha\beta(\alpha+\beta) + \gamma(\alpha-\beta)^2 - 3\gamma^2(\alpha+\beta)\}\lambda^2\nu^2 \\ & - 2\alpha(\beta+\gamma)^2\lambda^2\mu\nu \\ & + \beta(\beta+\gamma)^2\mu^4, \end{aligned}$$

and this is identically

$$\begin{aligned} &= \{(\alpha+\gamma)\lambda^2 + (\beta+\gamma)\mu^2 + 2\gamma\lambda\mu\} \\ &\quad \times \{\alpha(\gamma+\alpha)\nu^2 - 2(\beta\gamma+\gamma\alpha+\alpha\beta)\lambda\mu + \beta(\beta+\gamma)\mu^2\}; \end{aligned}$$

where observing that

$$\alpha\lambda^2 + \beta\mu^2 + \gamma(P-\lambda-\mu)^2 = X,$$

we have the first factor

$$(\alpha+\gamma)\lambda^2 + (\beta+\gamma)\mu^2 + 2\gamma\lambda\mu = X - \gamma P^2 + 2\gamma P(\lambda+\mu),$$

and consequently

$$\begin{aligned} \lambda^2\nu^2(A-B) = \; & - 2(\beta+\gamma)^2\mu^2X + 4\beta(\gamma+\alpha)^2\lambda\nu^2P \\ & + 2\{X - \gamma P^2 + 2\gamma P(\lambda+\mu)\}\{\alpha(\gamma+\alpha)\nu^2 - 2(\beta\gamma+\gamma\alpha+\alpha\beta)\lambda\mu + \beta(\beta+\gamma)\mu^2\}; \end{aligned}$$

viz. in virtue of $P = 0$, $X = 0$, we have $A = B$. And thus

$$A = B = C = -A_1 = -B_1 = -C_1;$$

so that the only values of Ω are, say, A and $-A$.

Reverting to the original equations

$$(f\theta^2+g)(k\theta y-x)^2+h(k-1)^2\theta^2z^2=0,$$

$$(bk^2\theta^2-a)(\theta y-x)^2+h(k-1)^2\theta^2w^2=0,$$

say these are

$$(\mathfrak{a},\mathfrak{b},\mathfrak{c},\mathfrak{d},\mathfrak{e})(\theta,1)^4=0,$$

$$(\mathfrak{a}',\mathfrak{b}',\mathfrak{c}',\mathfrak{d}',\mathfrak{e}')(\theta,1)^4=0,$$

then the coefficients in the two equations have the values

$$\begin{array}{ll}
fky^2, & bk^2y^2,\\
-2kfxy, & -2bk^2xy,\\
fx^2+gky^2+h(k-1)^2z^2, & bk^2x^2-ay^2+h(k-1)^2w^2,\\
-2gkxy, & 2axy,\\
gx^2, & -ax^2,
\end{array}$$

where observe that only $\mathfrak{c}$ contains z^2, and only $\mathfrak{c}'$ contains w^2. The result of the elimination is

$$\begin{vmatrix}
\mathfrak{a} & \mathfrak{b} & \mathfrak{c} & \mathfrak{d} & \mathfrak{e} & \cdot & \cdot & \cdot\\
\cdot & \mathfrak{a} & \mathfrak{b} & \mathfrak{c} & \mathfrak{d} & \mathfrak{e} & \cdot & \cdot\\
\cdot & \cdot & \mathfrak{a} & \mathfrak{b} & \mathfrak{c} & \mathfrak{d} & \mathfrak{e} & \cdot\\
\cdot & \cdot & \cdot & \mathfrak{a} & \mathfrak{b} & \mathfrak{c} & \mathfrak{d} & \mathfrak{e}\\
\mathfrak{a}' & \mathfrak{b}' & \mathfrak{c}' & \mathfrak{d}' & \mathfrak{e}' & \cdot & \cdot & \cdot\\
\cdot & \mathfrak{a}' & \mathfrak{b}' & \mathfrak{c}' & \mathfrak{d}' & \mathfrak{e}' & \cdot & \cdot\\
\cdot & \cdot & \mathfrak{a}' & \mathfrak{b}' & \mathfrak{c}' & \mathfrak{d}' & \mathfrak{e}' & \cdot\\
\cdot & \cdot & \cdot & \mathfrak{a}' & \mathfrak{b}' & \mathfrak{c}' & \mathfrak{d}' & \mathfrak{e}'
\end{vmatrix}=0;$$

viz. here the only terms which contain z^2 and w^2 are

$$\mathfrak{c}^4z^8\cdot\mathfrak{c}'^4w^8,$$

and hence the terms in z^8 and w^8 are

$$h^4(k-1)^8z^8\cdot a^4b^4k^8x^8y^8+h^4(k-1)^8w^8\cdot f^4g^4k^4x^8y^8,$$

viz. these are

$$=h^4k^8(k-1)^8x^8y^8\,(a^4b^4z^8+f^4g^4w^8),$$

or assuming that the determinant contains as a factor the function $b^2c^2f^2x^4+\cdots+2\Omega\,x^2y^2z^2w^2$, with a properly determined value of Ω, we see that the other factor is $=h^4k^8(k-1)^8x^4y^4$, which agrees with a preceding result.

[655] A Memoir on Differential Equations.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 292–339.]

We have to do with a set of variables, which is either unipartite $(x, y, z, \ldots)$, or else bipartite $(x, y, z, \ldots; p, q, r, \ldots)$, the variables in the latter case corresponding in pairs x and p, y and q, &c.

A letter not otherwise explained denotes a function of the variables. Any such letter may be put $=$ const., viz. we thereby establish a relation between the variables; and when this is so, we use the same letter to denote the constant value of the function. Thus the set being $(x, y, z; p, q, r)$, H may denote a given function $pqr - xyz$; and then, if $H =$ const., we have $pqr - xyz = H$ (a constant). This notation, when once clearly understood, is I think a very convenient one.

The present memoir relates chiefly to the following subjects:

A. *Unipartite set* $(x, y, z, \ldots)$. The differential system

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \cdots ,$$

and connected therewith the linear partial differential equation

$$X\frac{d\theta}{dx} + Y\frac{d\theta}{dy} + Z\frac{d\theta}{dz} + \cdots = 0 :$$

also the lineo-differential

$$X\,dx + Y\,dy + Z\,dz + \cdots .$$

B. *Bipartite set* $(x, y, z, \ldots; p, q, r, \ldots)$. The Hamiltonian system

$$\frac{dx}{\frac{dH}{dp}} = \frac{dy}{\frac{dH}{dq}} = \frac{dz}{\frac{dH}{dr}} = \cdots = \frac{dp}{-\frac{dH}{dx}} = \frac{dq}{-\frac{dH}{dy}} = \frac{dr}{-\frac{dH}{dz}} = \cdots$$

and connected therewith the linear partial differential equation

$$\frac{dH}{dp}\frac{d\theta}{dx}-\frac{dH}{dx}\frac{d\theta}{dp}+\frac{dH}{dq}\frac{d\theta}{dy}-\frac{dH}{dy}\frac{d\theta}{dq}+\cdots=0,$$

otherwise written

$$\frac{d(H,\theta)}{d(p,x)}+\frac{d(H,\theta)}{d(q,y)}+\cdots=0,$$

where H denotes a given function of the variables: also the Hamiltonian system as augmented by an equality $= dt$, and as augmented by this and another equality

$$= dV \div \left(p\,\frac{dH}{dp}+q\,\frac{dH}{dq}+r\,\frac{dH}{dr}+\cdots\right).$$

C. *Bipartite set* $(x,y,z,\ldots;p,q,r,\ldots)$. The partial differential equation $H=$ const., where, as before, H is a given function of the variables, but $p,q,r,\ldots$ are now the differential coefficients in regard to $x,y,z,\ldots$ respectively of a function V of these variables, or, what is the same thing, there exists a function

$$V=\int(p\,dx+q\,dy+r\,dz+\cdots),$$

of the variables $x,y,z,\ldots$.

In what precedes, I have written $(x,y,z,\ldots)$ to denote a set of any number n of variables, and $(x,y,z,\ldots;p,q,r,\ldots)$ to denote a set of any even number $2n$ of variables, and the investigations are for the most part applicable to these general cases. But for greater clearness and facility of expression, I usually consider the case of a set (x,y,z,w), or $(x,y,z;p,q,r)$, &c., as the case may be, consisting of a definite number of variables.

The greater part of the theory is not new, but I think that I have presented it in a more compact and intelligible form than has hitherto been done, and I have added some new results.

Introductory Remarks. Art. Nos. 1 to 3.

1. As already noticed, a letter not otherwise explained is considered as denoting a function of the variables of the set; but when necessary we indicate the variables by a notation such as $z=z(x,y)$; z is here a function (known or unknown as the case may be) of the variables x,y, the z on the right-hand side being in fact a functional symbol. And thus also $z=z(x,y),=$ const., denotes that the function $z(x,y)$ of the variables x,y has a constant value, which constant value is $=z$, viz. we thus indicate a relation between the variables x,y.

2. The variables x,y, &c., may have infinitesimal increments dx,dy, &c.; and the equations of connexion between the variables then give rise to linear relations between these increments, the coefficients therein being differential coefficients and,

as such, represented in the usual notation; thus if $z = z(x, y)$, we have $dz = \frac{dz}{dx}\,dx + \frac{dz}{dy}\,dy$, where $\frac{dz}{dx}, \frac{dz}{dy}$ are the so-called partial differential coefficients of z in regard to x, y respectively. If we have $y = y(x)$, then also $dy = \frac{dy}{dx}\,dx$, and the foregoing equation becomes

$$dz = \left(\frac{dz}{dx} + \frac{dz}{dy}\frac{dy}{dx}\right) dx;$$

but considering the two equations $z = z(x, y)$ and $y = y(x)$ as determining z as a function of x, say $z = z(x)$, we have $dz = \frac{d(z)}{dx}\,dx$; whence comparing the two formulae

$$\frac{d(z)}{dx} = \frac{dz}{dx} + \frac{dz}{dy}\frac{dy}{dx},$$

where $\frac{d(z)}{dx}$ is the so-called total differential coefficient of z in regard to x. The distinction is best made, not by any difference of notation $\left(\frac{d(z)}{dx}, \frac{dz}{dx}\right)$, but by appending in any case of doubt the equations or equation used in the differentiation. Thus we have $\frac{dz}{dx}$, where $z = z(x, y)$: or, as the case may be, $\frac{dz}{dx}$ where $z = z(x, y)$ and $y = y(x)$.

3. A relation between increments is always really a relation between differential coefficients; but we use the increments for symmetry and conciseness, as in the case of a differential system $\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z}$, or in a question relating to the lineo-differential $X\,dx + Y\,dy + Z\,dz$, for instance in the question whether this can be put $= du$.

Notations. Art. Nos. 4 to 6.

4. *Functional determinants.* If $a, b, c, \ldots$ are functions of the variables $x, y, z, w, \ldots$, then the determinants

$$\begin{vmatrix} \frac{da}{dx} & \frac{da}{dy} \\ \frac{db}{dx} & \frac{db}{dy} \end{vmatrix}, \qquad \begin{vmatrix} \frac{da}{dx} & \frac{da}{dy} & \frac{da}{dz} \\ \frac{db}{dx} & \frac{db}{dy} & \frac{db}{dz} \\ \frac{dc}{dx} & \frac{dc}{dy} & \frac{dc}{dz} \end{vmatrix}, \qquad \&c.,$$

are for shortness represented by

$$\frac{d(a, b)}{d(x, y)}, \qquad \frac{d(a, b, c)}{d(x, y, z)}, \qquad \&c.,$$

the notation being especially used in the first-mentioned case where the symbol is $\dfrac{d(a,b)}{d(x,y)} = 0$. It is sometimes convenient to extend this notation, and for instance use $\dfrac{d(a,b)}{d(x,y,z)}$ to denote the series of determinants

$$\begin{vmatrix} \dfrac{da}{dx} & \dfrac{da}{dy} & \dfrac{da}{dz} \\ \dfrac{db}{dx} & \dfrac{db}{dy} & \dfrac{db}{dz} \end{vmatrix}$$

which can be formed by selecting in every way two columns to form thereout a determinant; the equation

$$\frac{d(a,b)}{d(x,y,z)} = 0$$

will then denote that each of these determinants is $= 0$.

The analogous notation

$$\frac{d(a,b,c)}{d(x,y)}$$

would denote non-existent determinants, viz. there are here not columns enough to form with them a determinant: and the notation is not required.

5. In the case of a bipartite set $(x, y, z, \ldots; p, q, r, \ldots)$, if a, b are any functions of these variables, we consider the derivative

$$(a,b) = \frac{d(a,b)}{d(p,x)} + \frac{d(a,b)}{d(q,y)} + \frac{d(a,b)}{d(r,z)} + \cdots ,$$

viz. (a, b) is used to denote the sum of the functional determinants on the right hand.

6. Taking again $(x, y, z, w, \ldots)$ as the variables, then in the theory of the lineo-differential $X\,dx + Y\,dy + Z\,dz + W\,dw + \cdots$, we use certain derivative functions analogous to Pfaffians. They may be thus defined; viz. considering the numbers $1, 2, 3, 4, \ldots$ as corresponding to the variables $x, y, z, w, \ldots$ respectively, we have

$$1 = X, \quad 2 = Y, \quad 3 = Z, \quad 4 = W, \ \&c.,$$

$$12 = \frac{dY}{dx} - \frac{dX}{dy}, \qquad 13 = \frac{dZ}{dx} - \frac{dX}{dz}, \quad \&c.,$$

$$123 = 1 \cdot 23 + 2 \cdot 31 + 3 \cdot 12$$

$$= X\left(\frac{dZ}{dy} - \frac{dY}{dz}\right) + Y\left(\frac{dX}{dz} - \frac{dZ}{dx}\right) + Z\left(\frac{dY}{dx} - \frac{dX}{dy}\right),$$

$$1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23$$

$$= \left(\frac{dY}{dx} - \frac{dX}{dy}\right)\left(\frac{dW}{dz} - \frac{dZ}{dw}\right) + \left(\frac{dZ}{dx} - \frac{dX}{dz}\right)\left(\frac{dY}{dw} - \frac{dW}{dy}\right) + \left(\frac{dW}{dx} - \frac{dX}{dw}\right)\left(\frac{dZ}{dy} - \frac{dY}{dz}\right),$$

and, adding for greater distinctness the next following cases,

$$12345 = 1 \cdot 2345 + 2 \cdot 3451 + 3 \cdot 4512 + 4 \cdot 5123 + 5 \cdot 1234,$$

$$123456 = 12 \cdot 3456 + 13 \cdot 4561 + 14 \cdot 5612 + 15 \cdot 6123 + 16 \cdot 2345,$$

where of course 2345, &c., have the significations mentioned above.

Dependency of Functions. Art. Nos. 7 and 8.

7. Two or more functions of the same variables may be independent, or else dependent or connected; viz. in the latter case any one of the functions is a function of the others: $a = a(x)$, $b = b(x)$, the functions a, b are dependent, but if

$$a = a(x, y), \qquad b = b(x, y),$$

then the condition of dependency is

$$\frac{d(a, b)}{d(x, y)} = 0,$$

and, similarly, if $a = a(x, y, z)$, $b = b(x, y, z)$, then the conditions of dependency are

$$\frac{d(a, b)}{d(x, y, z)} = 0,$$

viz. if the equations thus represented are all of them satisfied, the functions are dependent, but if not, then they are independent.

Observe that, when $a = a(x, y, z)$, $b = b(x, y, z)$ as above, if we choose to attend only to the variables x, y, treating z as a mere constant, there is then a single condition of dependency $\dfrac{d(a, b)}{d(x, y)} = 0$, and so if we attend only to the variable x, treating y, z as mere constants, then a and b are dependent. Thus when $a = x$, $b = x^2 + y$, the functions a, b are independent if we attend to both the variables x, y; dependent if y be regarded as a constant.

8. Further when $a = a(x, y)$, $b = b(x, y)$, $c = c(x, y)$, the functions a, b, c are dependent; but when $a = a(x, y, z)$, $b = b(x, y, z)$, $c = c(x, y, z)$, the condition of dependency is

$$\frac{d(a, b, c)}{d(x, y, z)} = 0,$$

and so when $a = a(x, y, z, w)$, $b = b(x, y, z, w)$, $c = c(x, y, z, w)$, the conditions of dependency are

$$\frac{d(a, b, c)}{d(x, y, z, w)} = 0,$$

viz. if all the equations thus represented are satisfied, the functions are dependent; but if not, then they are independent. And so in other cases.

The General Differential System. Art. Nos. 9 to 22.

9. Taking the set of variables to be (x, y, z, w), the system is

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \frac{dw}{W},$$

and we associate with this the linear partial differential equation

$$X\frac{d\theta}{dx} + Y\frac{d\theta}{dy} + Z\frac{d\theta}{dz} + W\frac{d\theta}{dw} = 0.$$

10. It is tolerably evident that the differential equations establish between x, y, z, w a threefold relation depending upon three arbitrary constants; in fact, regarding (x, y, z, w) as the coordinates of a point in four-dimensional space, and starting from any given point, the differential equations determine the ratios of the increments dx, dy, dz, dw, that is, the direction of passage to a consecutive point; and then again taking for x, y, z, w the coordinates of this point, the same equations give the direction of passage to the next consecutive point, and so on. The locus of the point is therefore a curve, or we have between the coordinates a threefold relation, and (the initial point being arbitrary) we have a curve of the system through each point of the four-dimensional space, viz. the relation must involve three arbitrary constants. But this being so, the constants will be expressible as functions of the coordinates, viz. the threefold relation involving the three constants will be expressible in the form $a =$ const., $b =$ const., $c =$ const., where a, b, c denote respectively functions of the coordinates (x, y, z, w).

11. Supposing that one of the relations is $a =$ const., it is clear that the increment

$$da = \frac{da}{dx}\,dx + \frac{da}{dy}\,dy + \frac{da}{dz}\,dz + \frac{da}{dw}\,dw,$$

must become $= 0$, on substituting therein for dx, dy, dz, dw, the values X, Y, Z, W to which by virtue of the differential equations they are proportional, viz. that we must have identically

$$X\frac{da}{dx} + Y\frac{da}{dy} + Z\frac{da}{dz} + W\frac{da}{dw} = 0.$$

Conversely, when this is so, we have $da = 0$, by virtue of the differential equation.

We say that a is a solution of the partial differential equation, and an integral of the differential equations, viz. any solution of the partial differential equation is an integral of the differential equations, and any integral of the differential equations is a solution of the partial differential equation, or, this being so, we may in general without risk of ambiguity, say simply a is an integral[1]; similarly b and c are integrals, and, by what precedes, there are three integrals a, b, c.

[1]Viz. we use indifferently, in regard to the differential equations and to the partial differential equation, the term integral, which is appropriate to the differential equations; the appropriate term in regard to the partial differential equation would be solution.

Observe that, in speaking of an integral a, we mean a function of the variables; the differential equations give between the variables the relation $a = \text{const.}$, and when this is so, we use the same letter a to denote the constant value of this function.

12. In speaking of the three integrals a, b, c we mean independent integrals; any function whatever ϕa of an integral a, or any function whatever $\phi(a, b)$ of two integrals a, b, is an integral (viz. it is an integral of the differential equations, and also a solution of the partial differential equation), but such dependent integrals give nothing new, and we require a third independent integral c, viz. we need this to express the threefold relation between the variables, given by the differential equations, and also to express the general solution $\phi(a, b, c)$ of the partial differential equation.

13. By what precedes the analytical condition, in order that the integrals a, b, c may be independent, is that they are such as not to satisfy the relations

$$\frac{d(a, b, c)}{d(x, y, z, w)} = 0.$$

14. We moreover see *à posteriori*, that there cannot be more than three independent integrals; in fact, if a, b, c, d are integrals, then, considering them as solutions of the partial differential equation, we have four equations which by the elimination therefrom of X, Y, Z, W, give

$$\frac{d(a, b, c, d)}{d(x, y, z, w)} = 0,$$

and this is the very equation which expresses that a, b, c, d are not independent.

15. The notion of the integrals may be arrived at somewhat differently thus: take a, b, c, d any functions of the variables, and write

$$A = X\frac{da}{dx} + Y\frac{da}{dy} + Z\frac{da}{dz} + W\frac{da}{dw},$$

and the like for B, C, D: then replacing the original variables x, y, z, w by the new variables a, b, c, d, the differential equations become

$$\frac{da}{A} = \frac{db}{B} = \frac{dc}{C} = \frac{dd}{D},$$

where A, B, C, D are to be (by means of the given values of a, b, c, d as functions of x, y, z, w) expressed as functions of a, b, c, d. If then $A = 0$, $B = 0$, $C = 0$, the differential equations become

$$\frac{da}{0} = \frac{db}{0} = \frac{dc}{0} = \frac{dd}{D},$$

viz. we have $da = 0$, $db = 0$, $dc = 0$, and therefore $a = \text{const.}$, $b = \text{const.}$, $c = \text{const.}$, that is, we have the integrals a, b, c as before.

16. There is no general process for obtaining an integral a of the differential equations. Supposing such integral known, we can introduce it as a variable, in place of one of the original variables, say w, viz. we thus reduce the system to

$$\frac{dx}{X}=\frac{dy}{Y}=\frac{dz}{Z}=\frac{da}{0},$$

where X, Y, Z now denote the values assumed by these functions upon expressing therein w as a function of x, y, z, a, viz. they are now functions of x, y, z, a. The system thus breaks up into $da = 0$ and the system

$$\frac{dx}{X}=\frac{dy}{Y}=\frac{dz}{Z},$$

in which last (by virtue of the first equation, or $a =$ const.) a is to be regarded as a constant; the original system of three equations between four variables is thus reduced to a system of two equations between three variables. Supposing b to be an integral of this reduced system, b is given as a function of x, y, z, a, but upon substituting herein for a its value as a function of x, y, z, w, we have b a function of the original variables x, y, z, w, and b is then a second integral of the original system.

17. In like manner supposing a and b to be known, we reduce the system to the single equation

$$\frac{dx}{X}=\frac{dy}{Y},$$

where X, Y are now functions of x, y, a, b; supposing an integral hereof to be c, we have c a function of x, y, a, b; but upon substituting herein for a, b their values as functions of x, y, z, w, we have c a function of x, y, z, w, and as such it is the third integral of the original system.

18. It may be remarked that if, to the original system, we join on an equality $= dt$, viz. if we consider the system

$$\frac{dx}{X}=\frac{dy}{Y}=\frac{dz}{Z}=\frac{dw}{W}=dt,$$

where X, Y, Z, W are as before functions of the variables (x, y, z, w), then the integrals a, b, c of the original system being known, we can by means of them express for instance X as a function of x, a, b, c, and we have then, const. $= t - \int \frac{dx}{X}$, where the integration is to be performed regarding a, b, c as constants; writing $\int \frac{dx}{X} = \tau$, but after the integration replacing a, b, c by their values as functions of x, y, z, w, we have τ a function of x, y, z, w; and we say that $t - \tau$ is an integral; putting it $=$ const., we use also t to denote the constant value of the integral $t - \tau$ in question. Observe that here, the integrals a, b, c being known, the last integral $t - \tau$ is obtained by a quadrature.

19. The result would have been similar, if the adjoined equality had been $= \dfrac{dt}{T}$, (T a function of x, y, z, w), but in reference to subsequent matter, I retain the equality $= dt$, and adjoin a second equality $= \dfrac{dV}{\Omega}$ (Ω a function of x, y, z, w); we have then the integral $t - \tau$ as before, and another integral $V - \displaystyle\int \frac{\Omega\, dx}{X}$, where Ω, X are first expressed as functions of x, a, b, c, but after the integration a, b, c are replaced by their values as functions of (x, y, z, w), say this is the integral $V - \lambda$; this, when the integrals a, b, c are known, is (like $t - \tau$) obtained by a quadrature.

20. Attending only to the adjoined equality $= dt$, we can by means of the four integrals express each of the variables x, y, z, w as a function of $a, b, c, t - \tau$; viz. these four equations, regarding therein $t - \tau$ as a variable parameter, are in fact equivalent to the equations $a =$ const., $b =$ const., $c =$ const., which connect the variables x, y, z, w with the integrals a, b, c regarded as constants.

21. All that precedes is of course applicable to a system of $n - 1$ equations between n variables, the number of independent integrals being $= n - 1$.

22. I take an example with the three variables x, y, z; the differential equations being

$$\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)},$$

and therefore the partial differential equation

$$x(y-z)\frac{d\theta}{dx} + y(z-x)\frac{d\theta}{dy} + z(x-y)\frac{d\theta}{dz} = 0.$$

The integrals are $a = x + y + z$, $b = xyz$; and it will be shown how either of these integrals being known, the system is reduced to a single equation between two variables, say x, y.

First, a being known, $= x + y + z$ as before, we have

$$x(y-z) = x(x+2y-a), \qquad y(z-x) = y(a-2x-y),$$

and the system is

$$\frac{dx}{x(x+2y-a)} = \frac{dy}{y(a-2x-y)},$$

which has the integral $b = xy(a - x - y)$; observe that this is a solution of the partial differential equation

$$x(x+2y-a)\frac{d\theta}{dx} + y(a-2x-y)\frac{d\theta}{dy} = 0.$$

For a putting its value we find $b = xyz$.

Secondly, b being known, $= xyz$ as before, we have

$$x(y-z) = xy - \frac{b}{z}, \qquad y(z-x) = \frac{b}{x} - xy,$$

and the system is

$$\frac{dx}{xy - \dfrac{b}{y}} = \frac{dy}{\dfrac{b}{x} - xy},$$

which has the integral $a = x + y + \dfrac{b}{xy}$; observe that this is a solution of the partial differential equation

$$\left(xy - \frac{b}{y}\right)\frac{d\theta}{dx} + \left(\frac{b}{x} - xy\right)\frac{d\theta}{dy} = 0.$$

For b putting its value, we find $a = x + y + z$.

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23. First, if there are only two variables (x, y), the system consists of the single equation

$$\frac{dx}{X} = \frac{dy}{Y},$$

which may be written

$$Y\,dx - X\,dy = 0.$$

Hence, if a be an integral, we have

$$\frac{da}{dx}\,dx + \frac{da}{dy}\,dy = 0;$$

the two will agree if there exists a function M such that

$$\frac{da}{dx} = MY, \qquad \frac{da}{dy} = -MX,$$

and thence, in virtue of the identity

$$\frac{d}{dy}\frac{da}{dx} = \frac{d}{dx}\frac{da}{dy},$$

we find

$$\frac{d(MX)}{dx} + \frac{d(MY)}{dy} = 0,$$

or, as this may also be written,

$$X\frac{dM}{dx} + Y\frac{dM}{dy} + M\left(\frac{dX}{dx} + \frac{dY}{dy}\right) = 0,$$

as the condition to determine the multiplier M. Supposing M known, we have $M(Y\,dx - X\,dy) = da$, or say $a = \displaystyle\int M(Y\,dx - X\,dy)$, viz. the integral a is determined by a quadrature.

24. In the case of three variables (x, y, z), the system is

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z},$$

or, writing these in the form

$$Y\,dz - Z\,dy = 0, \quad Z\,dx - X\,dz = 0, \quad X\,dy - Y\,dx = 0,$$

the course which immediately suggests itself is to seek for factors L, M, N, such that, a being an integral, we may have

$$L(Y\,dz - Z\,dy) + M(Z\,dx - X\,dz) + N(X\,dy - Y\,dx) = da,$$

but this does not lead to any result. The course taken by Jacobi is quite a different one: he, in fact, determines a multiplier M connected with two integrals a, b.

25. Supposing that a, b are independent integrals, we have

$$X\frac{da}{dx} + Y\frac{da}{dy} + Z\frac{da}{dz} = 0,$$

$$X\frac{db}{dx} + Y\frac{db}{dy} + Z\frac{db}{dz} = 0;$$

and determining from these equations the ratios of the quantities X, Y, Z, we may, it is clear, write

$$MX, MY, MZ = \frac{d(a,b)}{d(y,z)}, \frac{d(a,b)}{d(z,x)}, \frac{d(a,b)}{d(x,y)}.$$

It may be shown that we have identically

$$\frac{d}{dx}\frac{d(a,b)}{d(y,z)} + \frac{d}{dy}\frac{d(a,b)}{d(z,x)} + \frac{d}{dz}\frac{d(a,b)}{d(x,y)} = 0,$$

and we thence deduce

$$\frac{d(MX)}{dx} + \frac{d(MY)}{dy} + \frac{d(MZ)}{dz} = 0,$$

or, what is the same thing,

$$X\frac{dM}{dx} + Y\frac{dM}{dy} + Z\frac{dM}{dz} + M\left(\frac{dX}{dx} + \frac{dY}{dy} + \frac{dZ}{dz}\right) = 0,$$

as the condition for determining the multiplier M.

26. The use is as follows: supposing that M is known, and supposing also that one integral a of the system is known, we can then by a quadrature determine the other integral b. Thus, supposing that we know the integral $a, = a(x, y, z)$, we can by means of this integral express z in terms of x, y, a; and hence we may regard the unknown integral b as expressed in the like form, $b = b(x, y, a)$. The original values of $\frac{db}{dx}, \frac{db}{dy}, \frac{db}{dz}$ become on this supposition

$$\frac{db}{dx} + \frac{db}{da}\frac{da}{dx}, \quad \frac{db}{dy} + \frac{db}{da}\frac{da}{dy}, \quad \frac{db}{da}\frac{da}{dz},$$

and we thence find

$$\frac{d(a,b)}{d(y,z)},\quad \frac{d(a,b)}{d(z,x)},\quad \frac{d(a,b)}{d(x,y)} = -\frac{da}{dz}\frac{db}{dy},\quad \frac{da}{dz}\frac{db}{dx},\quad \frac{d(a,b)}{d(x,y)}.$$

We have therefore

$$MX, MY = -\frac{da}{dz}\frac{db}{dy},\quad \frac{da}{dz}\frac{db}{dx},$$

and, consequently,

$$db = \frac{db}{dx}\,dx + \frac{db}{dy}\,dy = \frac{M}{\frac{da}{dz}}(Y\,dx - X\,dy);$$

viz. $M, \dfrac{da}{dz}, Y, X$ being all of them expressed as functions of x, y, a, the expression on the right-hand is a complete differential, and we have

$$b = \int \frac{M}{\frac{da}{dz}}(Y\,dx - X\,dy);$$

that is, the integral b is determined by a quadrature.

27. Thus, in the example No. 22,

$$\frac{dX}{dx} + \frac{dY}{dy} + \frac{dZ}{dz} = 0,$$

and a value of the multiplier is $= 1$. Supposing that the given integral is $a = x + y + z$, then $\dfrac{da}{dz} = 1$, and we have accordingly 1 as the multiplier of the equation

$$y(a - 2x - y)\,dx + x(a - x - 2y)\,dy = 0,$$

viz. this equation is integrable *per se*. Supposing the given integral to be $b = xyz$, then $\dfrac{db}{dz} = xy$, viz. we have $\dfrac{1}{xy}$ as the multiplier of the equation

$$\left(\frac{b}{x} - xy\right)dx + \left(xy - \frac{b}{y}\right)dy = 0,$$

and we thus in each case obtain the other integral as before.

28. The foregoing result may be presented in a more symmetrical form by taking in place of x, y any two variables $u = u(x, y, z)$, $v = v(x, y, z)$.

Supposing the integral a known as before, the system then is

$$\frac{du}{U} = \frac{dv}{V} = \frac{da}{0},$$

where $U, V = X\dfrac{du}{dx} + Y\dfrac{du}{dy} + Z\dfrac{du}{dz},\ X\dfrac{dv}{dx} + Y\dfrac{dv}{dy} + Z\dfrac{dv}{dz}$, these being expressed as functions of u, v, a; or, what is the same thing, we have $V\,du - U\,dv = 0$, a being in this equation regarded as a constant.

From the foregoing values of MX, MY, MZ, we deduce

$$MU, MV = \frac{d(u,a,b)}{d(x,y,z)}, \frac{d(v,a,b)}{d(x,y,z)}.$$

But forming the values of du, dv, da, db, we have an equation, determinant $=0$, which equation may be written

$$\frac{d(v,a,b)}{d(x,y,z)}\,du - \frac{d(u,a,b)}{d(x,y,z)}\,dv + \frac{d(u,v,b)}{d(x,y,z)}\,da - \frac{d(u,v,a)}{d(x,y,z)}\,db = 0,$$

or, writing herein $da = 0$, this is

$$\frac{d(v,a,b)}{d(x,y,z)}\,du - \frac{d(u,a,b)}{d(x,y,z)}\,dv = \frac{d(u,v,a)}{d(x,y,z)}\,db,$$

viz. this is

$$M(V\,du - U\,dv) = db\,\frac{d(u,v,a)}{d(x,y,z)},$$

or say

$$db = M(V\,du - U\,dv) \div \frac{d(u,v,a)}{d(x,y,z)},$$

where, on the right-hand side, everything must be expressed in terms of u, v, a. It thus appears that on expressing the final equation as a relation $V\,du - U\,dv = 0$ between the variables u and v, the multiplier hereof is $M \div \dfrac{d(u,v,a)}{d(x,y,z)}$. If $u, v = x, y$, this agrees with a foregoing result.

29. The theory is precisely the same for any number of variables. Thus, if there are four variables x, y, z, w, we have

$$MX, MY, MZ, MW = \frac{d(a,b,c)}{d(y,z,w)}, \frac{d(a,b,c)}{d(z,w,x)}, \frac{d(a,b,c)}{d(w,x,y)}, \frac{d(a,b,c)}{d(x,y,z)},$$

and, we have between the functions on the right-hand an identical relation, in virtue of which

$$\frac{d(MX)}{dx} + \frac{d(MY)}{dy} + \frac{d(MZ)}{dz} + \frac{d(MW)}{dw} = 0;$$

then, supposing that a value of M is known, and also any two integrals a, b, and that by means of these the equation to be finally integrated is expressed as a relation $V\,du - U\,dv = 0$ between any two variables u and v, the multiplier of this is

$$= M \div \frac{d(u,v,a,b)}{d(x,y,z,w)},$$

where U, V and this multiplier are to be expressed in terms of u, v, a, b.

The general result is that, given a value of the multiplier, and also all but one of the integrals, the final integral is expressible by a quadrature.